

```
import cv2
import numpy as np
from google.colab import files
from google.colab.patches import cv2_imshow

# Upload image
uploaded = files.upload()

# Get image name
filename = list(uploaded.keys())[0]

# Read image
img = cv2.imread(filename)

# Resize image for easy viewing
img = cv2.resize(img, (500, 400))

# Laplacian mask with diagonal neighbors
kernel = np.array([
    [1, 1, 1],
    [1, -8, 1],
    [1, 1, 1]
])

# Apply Laplacian filter
laplacian = cv2.filter2D(img, -1, kernel)

# Sharpen the image
sharpened = cv2.subtract(img, laplacian)

# Save output
cv2.imwrite("Sharpened_Image.jpg", sharpened)

# Display original image
print("Original Image:")
cv2_imshow(img)

# Display sharpened image
print("Sharpened Image:")
cv2_imshow(sharpened)

print("Output saved as Sharpened_Image.jpg")
```

Choose files img1.jpeg

img1.jpeg(image/jpeg) - 14202 bytes, last modified: 20/07/2026 - 100% done

Saving img1.jpeg to img1 (1).jpeg

Original Image:



Sharpened Image:



Output saved as Sharpened_Image.jpg

